Ethylene–air ZND validation

FFCM2 and the tuned reduced mechanism

12 cases at 298 K

Pressure: 1, 5 and 20 bar
Equivalence ratio: 0.5, 1.0, 1.5 and 2.0

Endpoint temperatures agree closely.
Reaction-zone positions and widths remain inaccurate.

Errors across all 12 conditions

Signed errors use 100 × (model/reference − 1). Temperature differences are model − reference.

Case	P (bar)	φ	ΔT (K)	Induction error	Exothermic error
1	1	0.5	+1.984	-38.06%	+35.45%
2	1	1	+5.909	-51.43%	-46.34%
3	1	1.5	-8.284	-79.18%	-38.87%
4	1	2	+2.283	-72.68%	-72.97%
5	5	0.5	+10.182	+24.67%	-45.26%
6	5	1	+23.741	-44.29%	-43.84%
7	5	1.5	-12.626	-58.42%	-81.52%
8	5	2	+52.994	-45.40%	-90.97%
9	20	0.5	+1.733	+8.98%	-53.44%
10	20	1	-5.111	-72.93%	-81.51%
11	20	1.5	-5.320	-67.14%	-87.87%
12	20	2	-1.029	-64.26%	-84.09%

Largest length discrepancy: case 8, exothermic length −90.97% (about 11.1× too short).

Comparison measures

Equal-case summaries describe this validation set, not the optimizer’s weighted loss.

$$\text{Signed percentage error} = 100 \times (q_{model} / q_{FFCM2} - 1)$$

$$\text{Absolute log error} = |\ln(\ell_{model} / \ell_{FFCM2})|$$

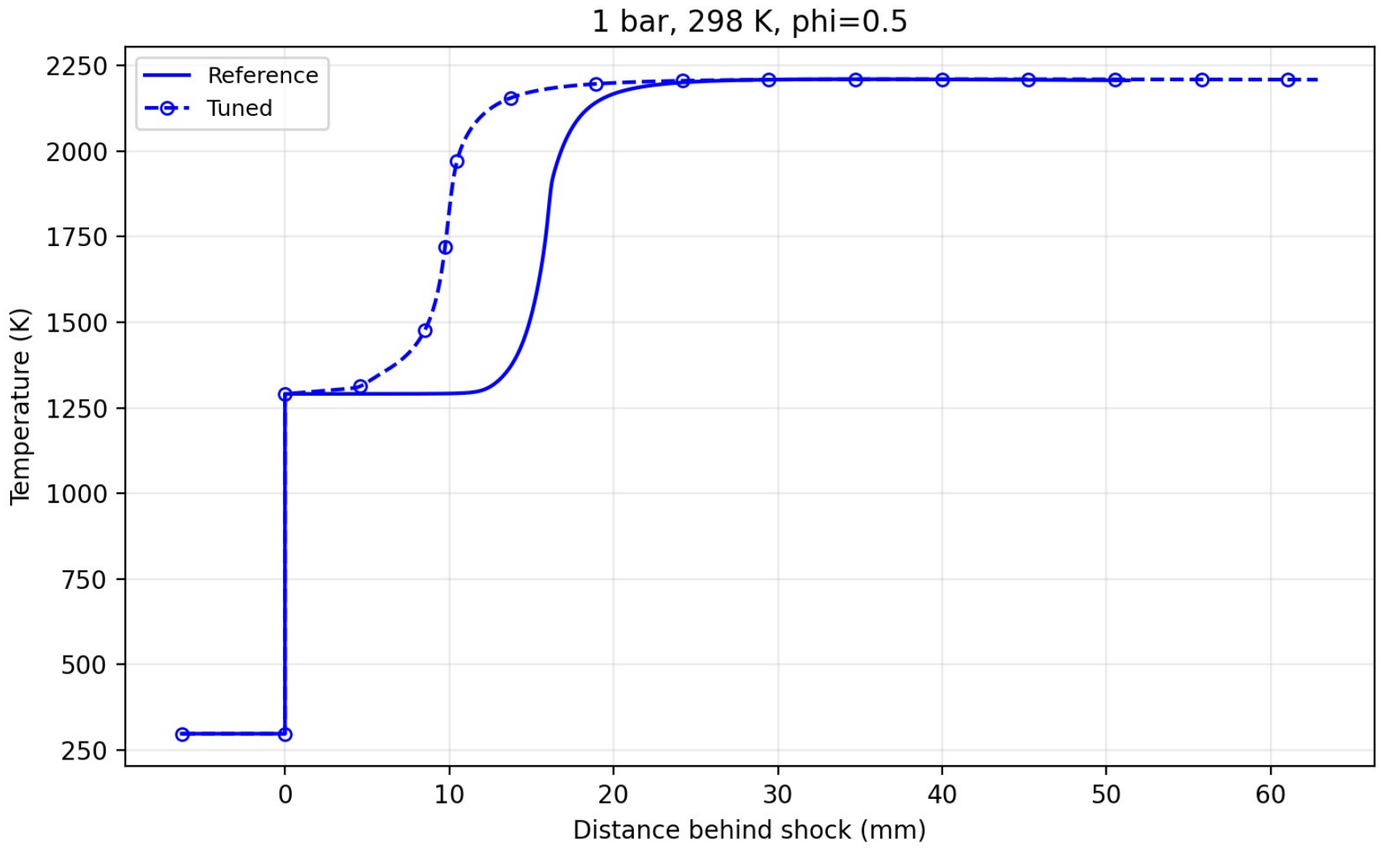
Zero means agreement. A log error of ln(2) ≈ 0.693 means a factor-of-two difference.

Metric	Mean absolute error	Largest absolute error
Final temperature	10.933 K (0.393%)	52.994 K (case 8)
Induction length	52.29%	79.18% (case 3)
Exothermic length	63.51%	90.97% (case 8)

Temperature RMSE: 17.81 K. Length values in the case tables are in metres.

Case 1: 1 bar, φ = 0.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2206.24	2208.23	+0.09%
ℓind (m)	1.6069e-2	9.9534e-3	-38.06%
ℓexo (m)	7.4218e-4	1.0052e-3	+35.45%

ΔT = +1.984 K

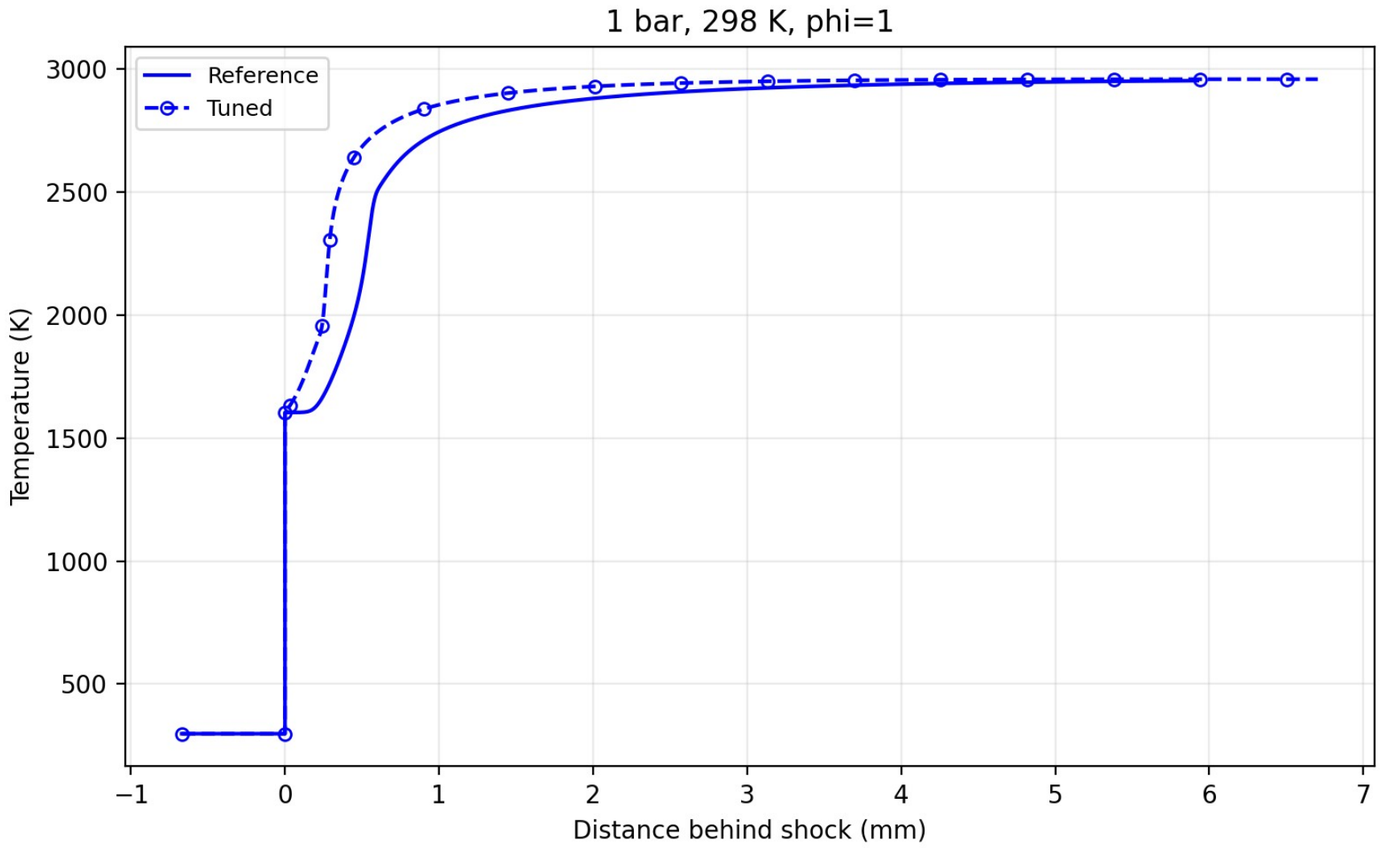
|ln(ℓind,model / ℓind,FFCM2)| = 0.478982

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.303396

The tuned temperature rise occurs earlier. Final temperature differs by only 1.984 K.

Case 2: 1 bar, φ = 1.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2952.20	2958.11	+0.20%
ℓind (m)	5.4732e-4	2.6583e-4	-51.43%
ℓexo (m)	1.1799e-4	6.3310e-5	-46.34%

ΔT = +5.909 K

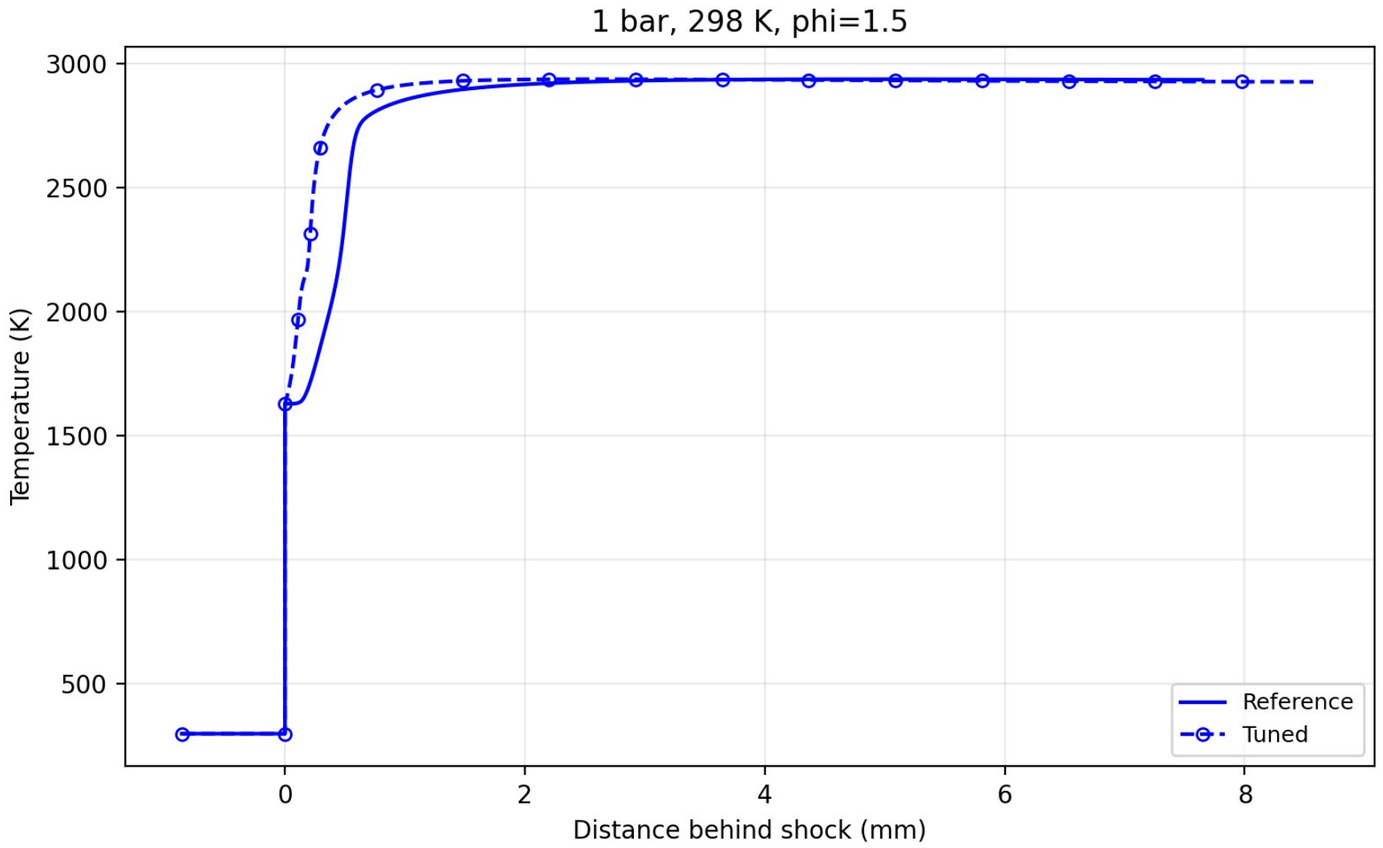
|ln(ℓind,model / ℓind,FFCM2)| = 0.722167

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.622549

Induction and exothermic lengths are approximately half the reference values.

Case 3: 1 bar, φ = 1.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2933.82	2925.53	-0.28%
ℓind (m)	5.2460e-4	1.0921e-4	-79.18%
ℓexo (m)	1.5716e-4	9.6069e-5	-38.87%

ΔT = -8.284 K

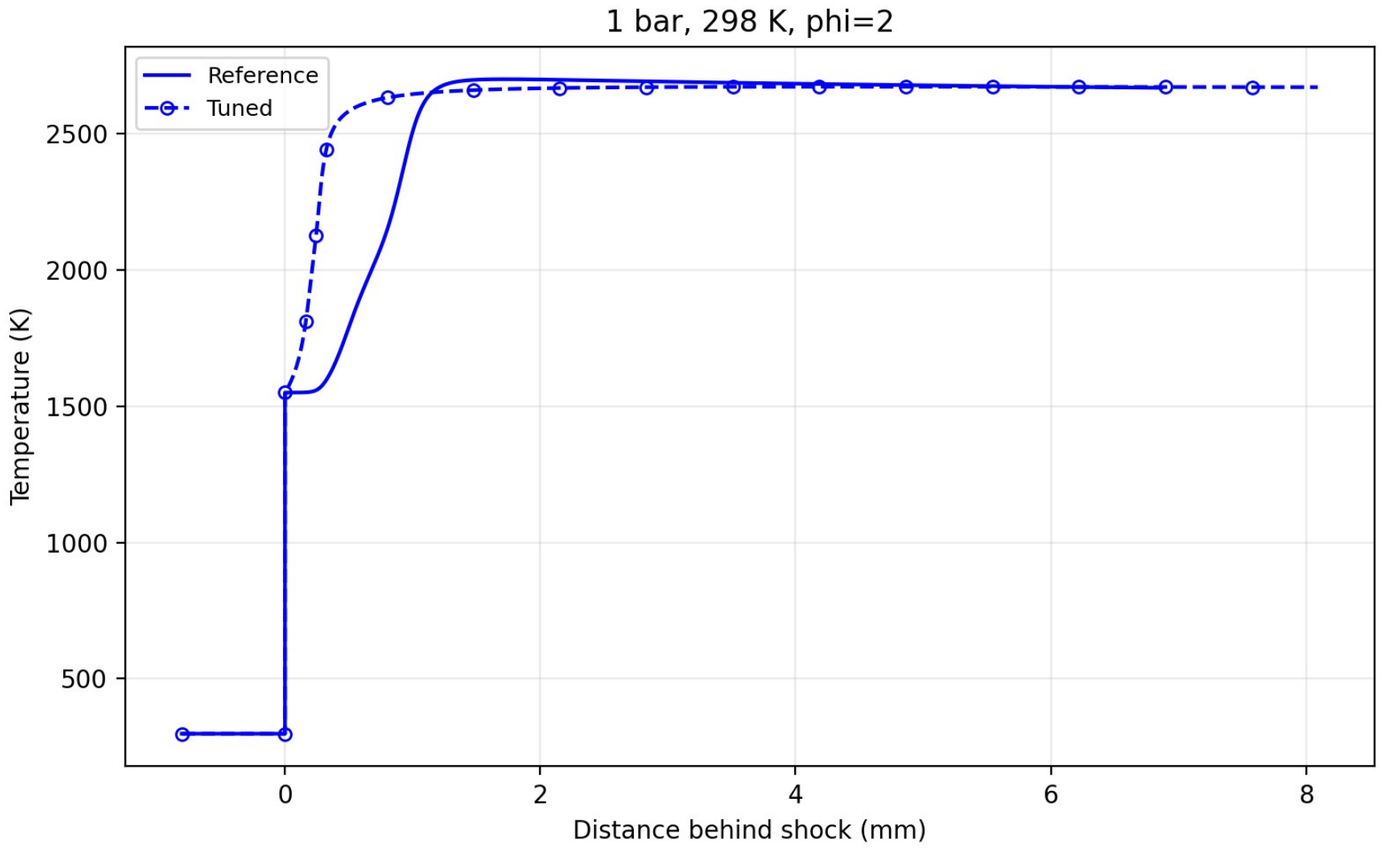
|ln(ℓind,model / ℓind,FFCM2)| = 1.56937

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.492183

The tuned induction length is 4.8× shorter than FFCM2 at φ = 1.5.

Case 4: 1 bar, φ = 2.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2668.92	2671.20	+0.09%
ℓind (m)	9.4947e-4	2.5943e-4	-72.68%
ℓexo (m)	6.2757e-4	1.6965e-4	-72.97%

ΔT = +2.283 K

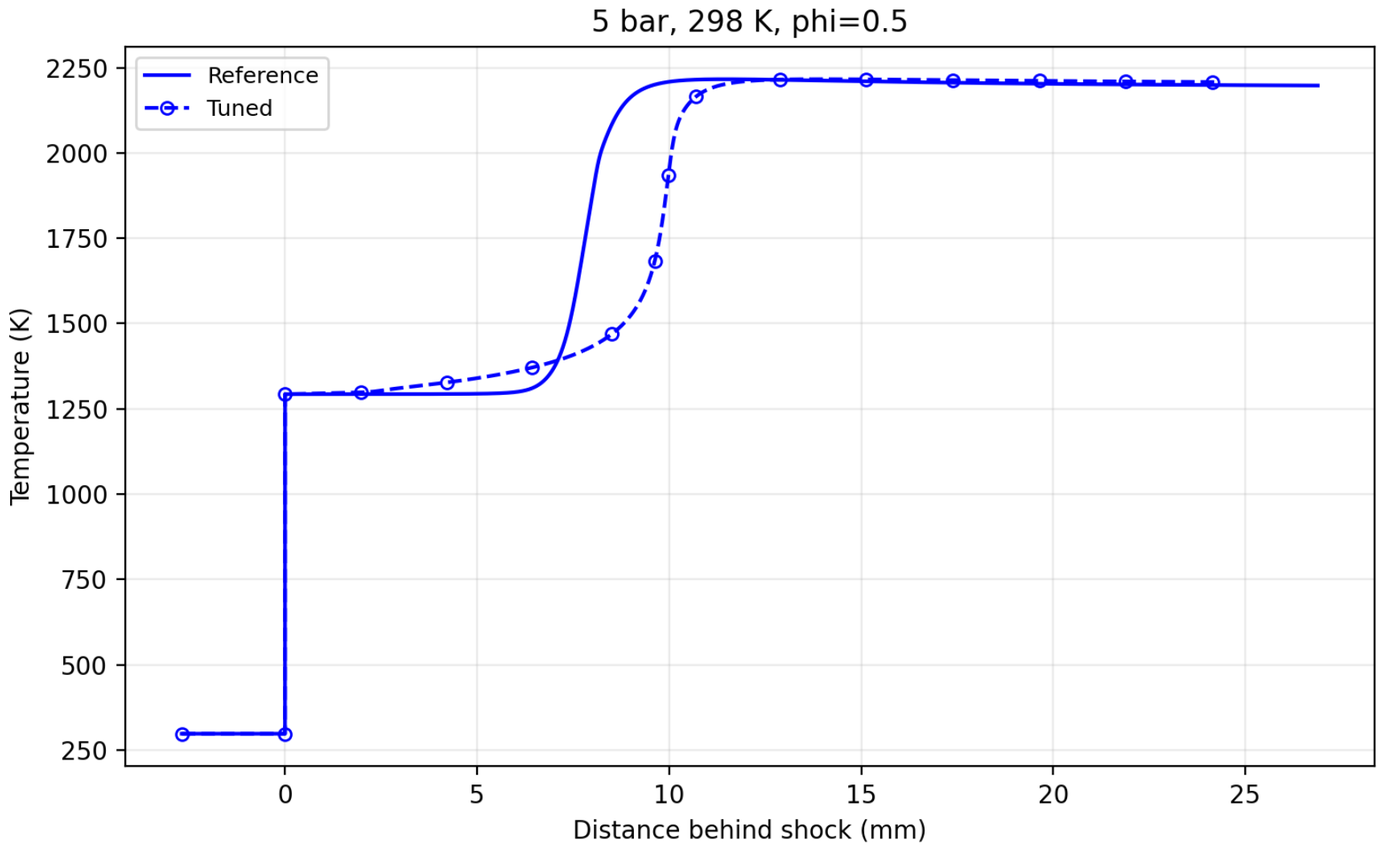
|ln(ℓind,model / ℓind,FFCM2)| = 1.2974

|ln(ℓexo,model / ℓexo,FFCM2)| = 1.30812

Both characteristic lengths are about 73% shorter than the reference.

Case 5: 5 bar, φ = 0.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2197.44	2207.62	+0.46%
ℓind (m)	8.0083e-3	9.9836e-3	+24.67%
ℓexo (m)	8.8781e-4	4.8595e-4	-45.26%

ΔT = +10.182 K

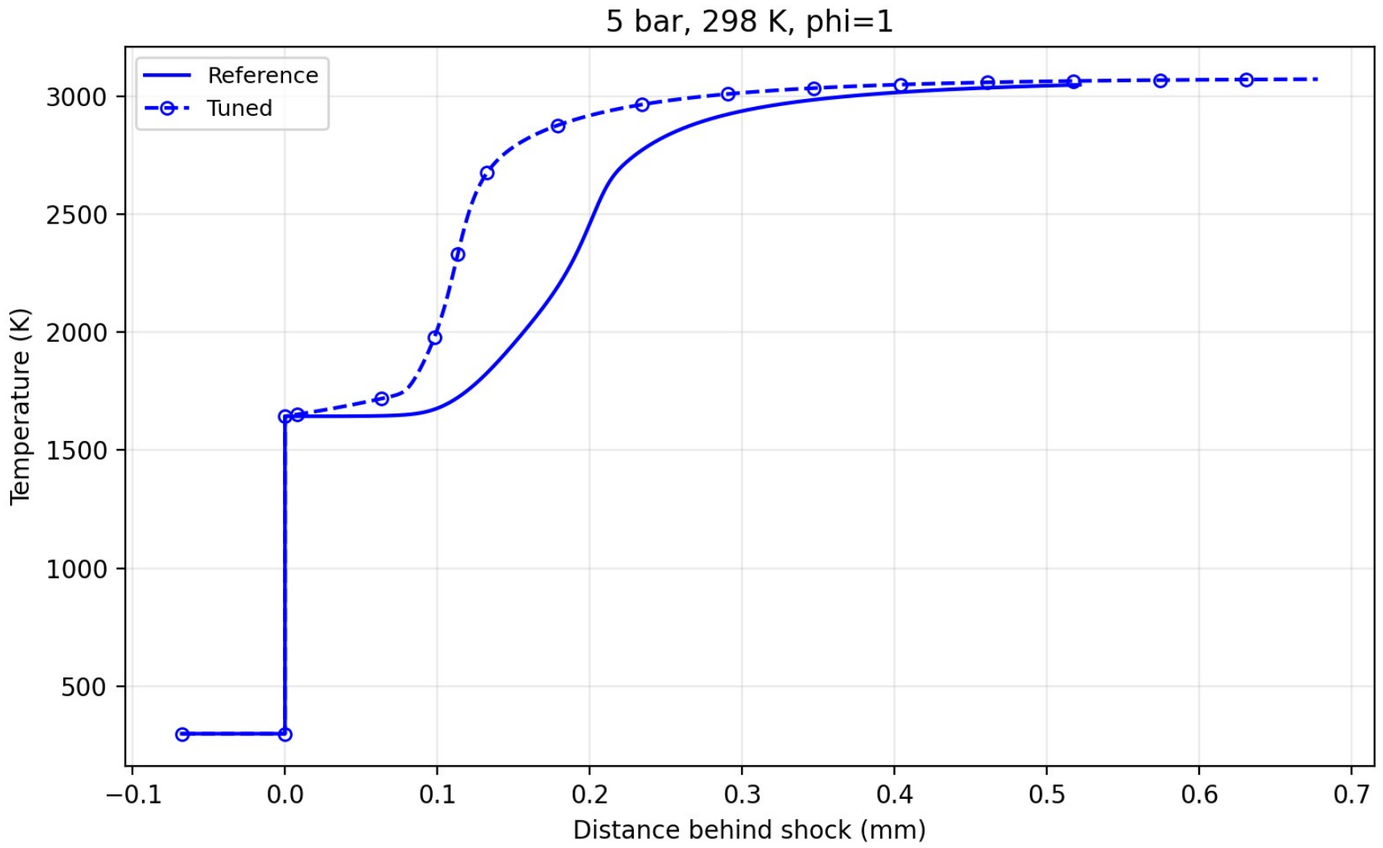
|ln(ℓind,model / ℓind,FFCM2)| = 0.220462

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.602653

Induction is 24.67% longer, while the exothermic zone is 45.26% shorter.

Case 6: 5 bar, φ = 1.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	3049.34	3073.08	+0.78%
ℓind (m)	2.0274e-4	1.1295e-4	-44.29%
ℓexo (m)	5.0900e-5	2.8587e-5	-43.84%

ΔT = +23.741 K

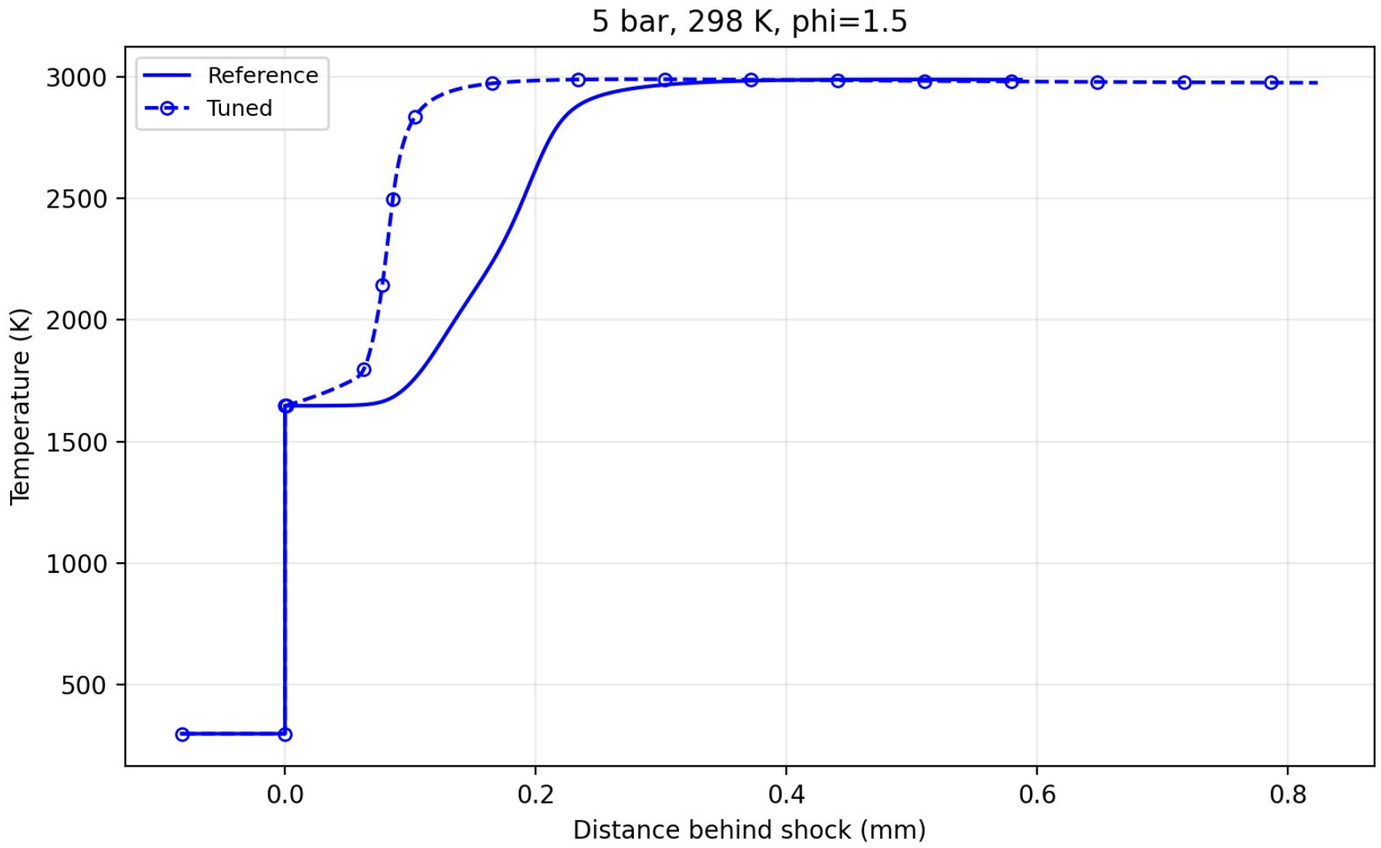
|ln(ℓind,model / ℓind,FFCM2)| = 0.584975

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.576891

The tuned profile rises earlier and more sharply than the reference profile.

Case 7: 5 bar, φ = 1.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2987.70	2975.08	-0.42%
ℓind (m)	2.0044e-4	8.3337e-5	-58.42%
ℓexo (m)	1.0433e-4	1.9279e-5	-81.52%

ΔT = -12.626 K

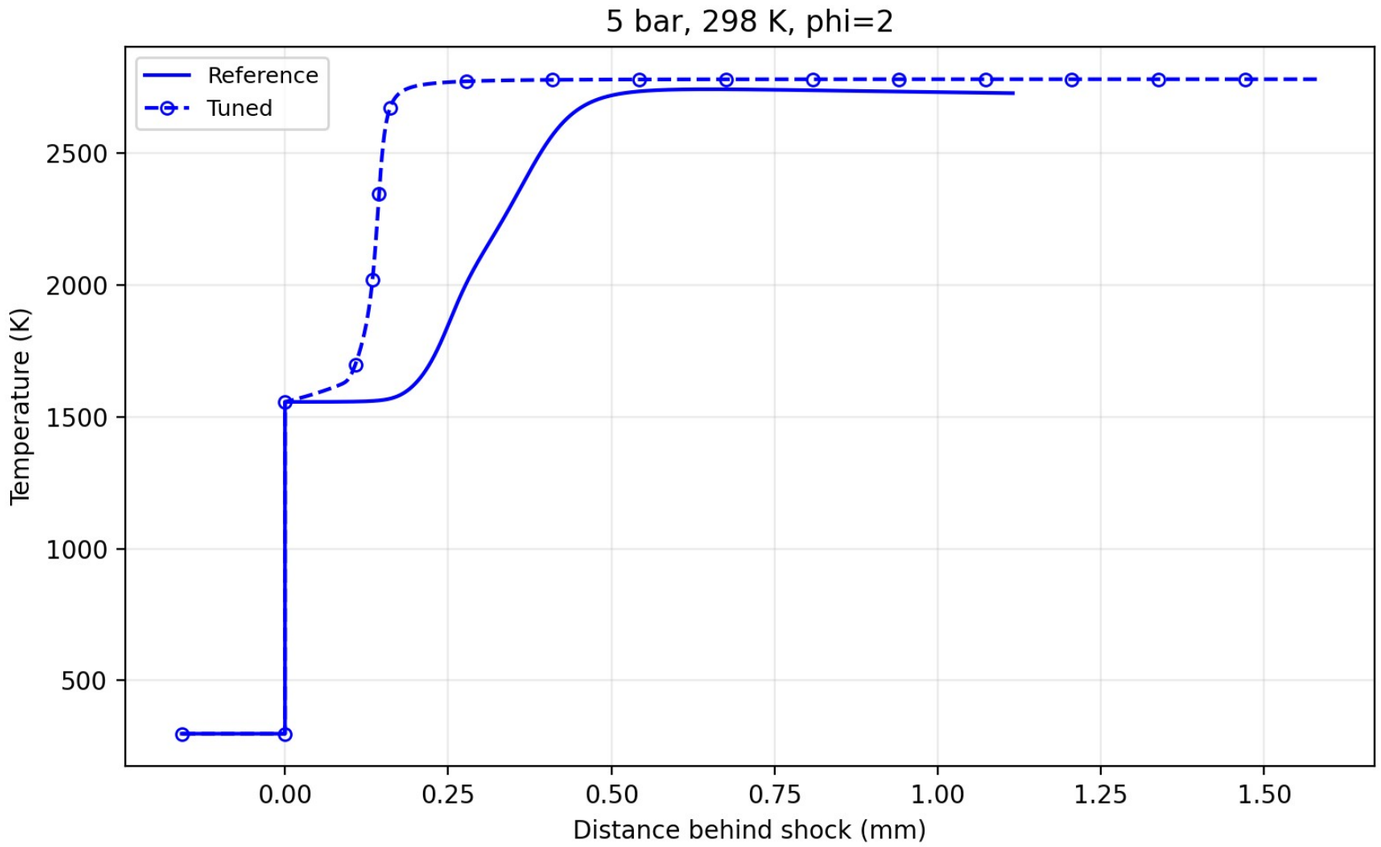
|ln(ℓind,model / ℓind,FFCM2)| = 0.877605

|ln(ℓexo,model / ℓexo,FFCM2)| = 1.68852

The exothermic length is 5.4× shorter, despite a small endpoint-temperature error.

Case 8: 5 bar, φ = 2.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2727.72	2780.71	+1.94%
ℓind (m)	2.6407e-4	1.4418e-4	-45.40%
ℓexo (m)	2.3135e-4	2.0891e-5	-90.97%

ΔT = +52.994 K

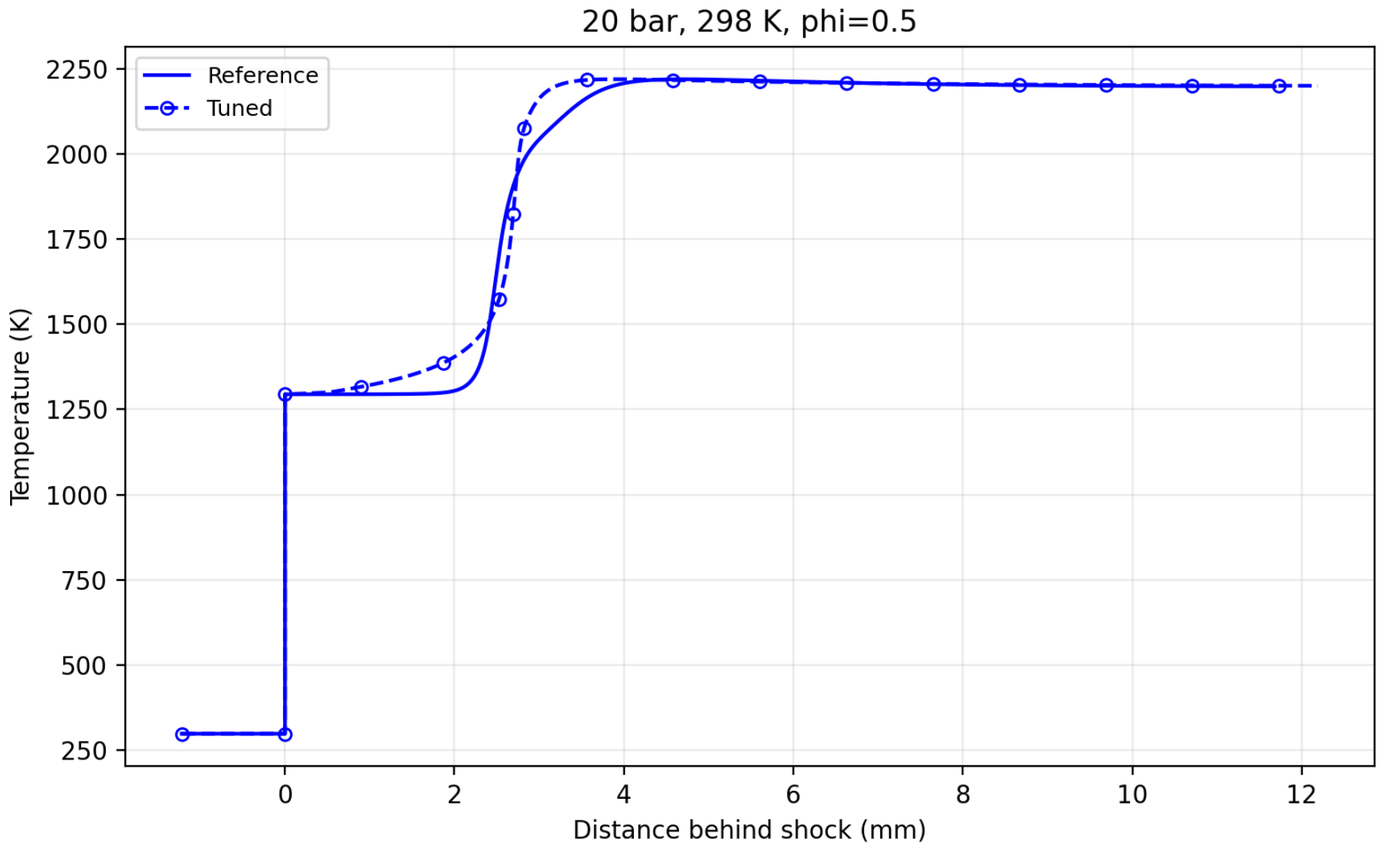
|ln(ℓind,model / ℓind,FFCM2)| = 0.605148

|ln(ℓexo,model / ℓexo,FFCM2)| = 2.4046

This case has the largest temperature and exothermic-length discrepancies.

Case 9: 20 bar, φ = 0.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2198.30	2200.03	+0.08%
ℓind (m)	2.4949e-3	2.7188e-3	+8.98%
ℓexo (m)	3.3413e-4	1.5556e-4	-53.44%

ΔT = +1.733 K

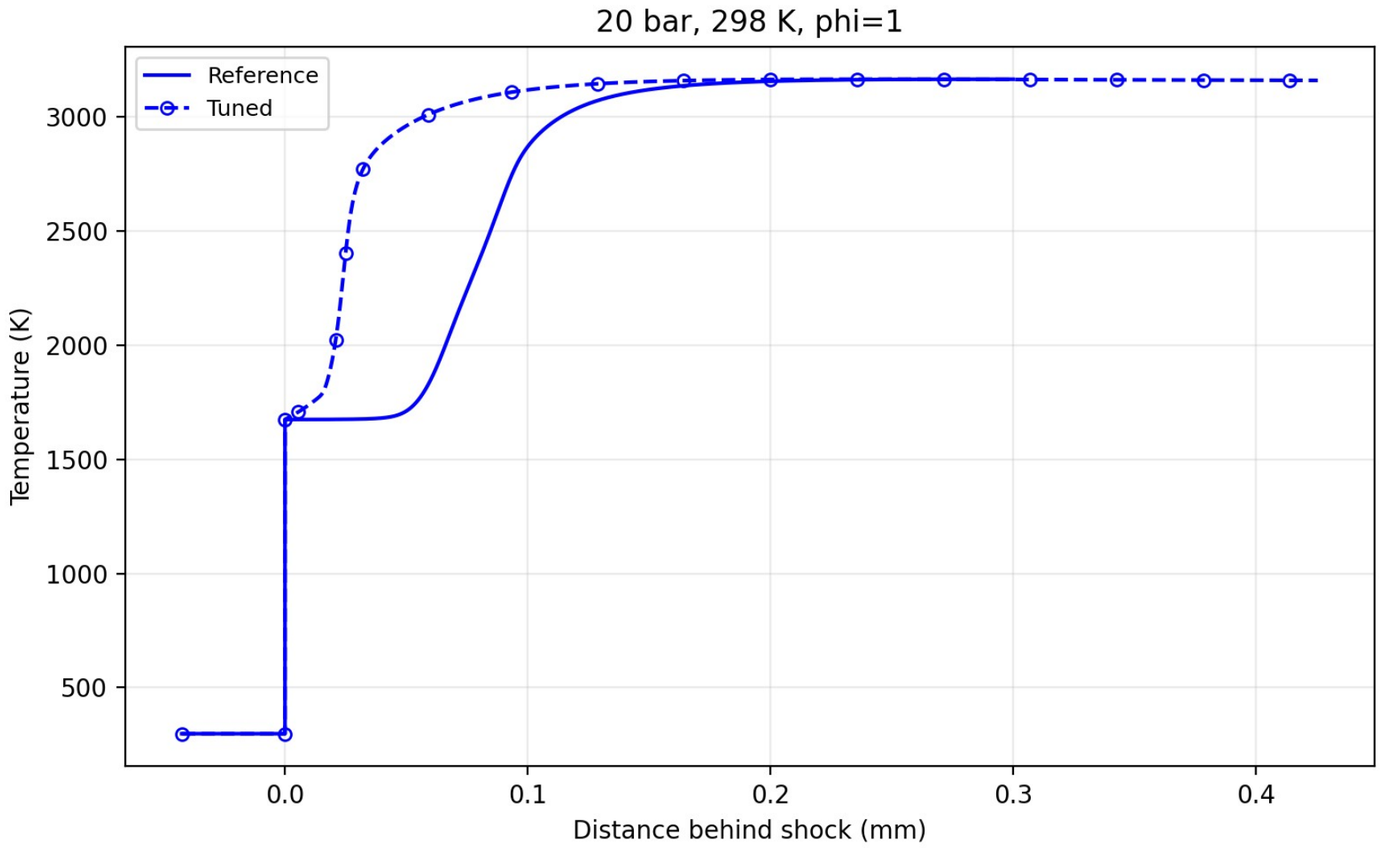
|ln(ℓind,model / ℓind,FFCM2)| = 0.0859669

|ln(ℓexo,model / ℓexo,FFCM2)| = 0.764486

Induction is within 8.98%, but the exothermic length remains 53.44% too short.

Case 10: 20 bar, φ = 1.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	3164.69	3159.58	-0.16%
ℓind (m)	8.8824e-5	2.4049e-5	-72.93%
ℓexo (m)	4.0357e-5	7.4619e-6	-81.51%

ΔT = -5.111 K

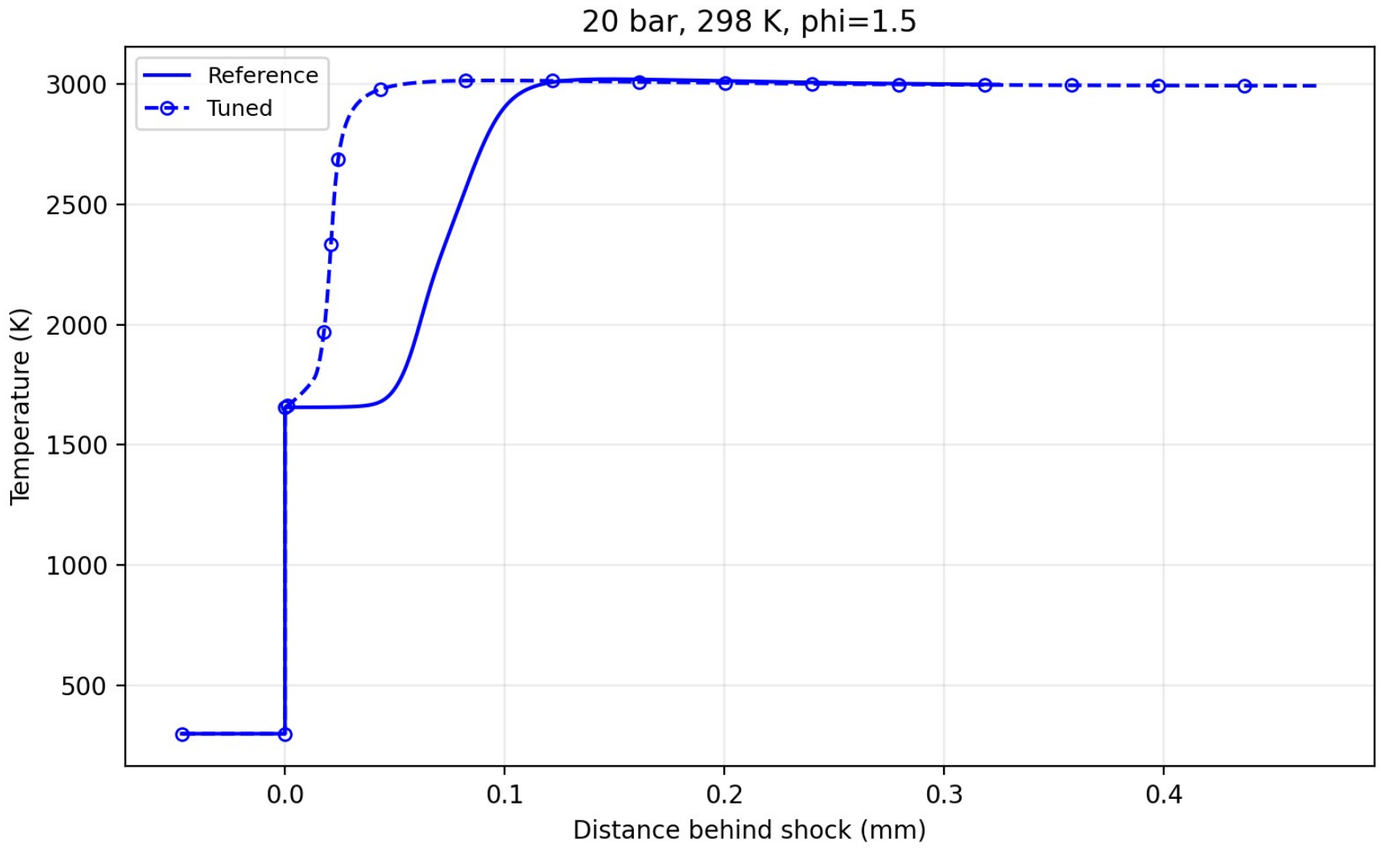
|ln(ℓind,model / ℓind,FFCM2)| = 1.30657

|ln(ℓexo,model / ℓexo,FFCM2)| = 1.68795

The reaction-zone lengths are much shorter, while the final temperature nearly matches.

Case 11: 20 bar, φ = 1.5

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2998.07	2992.75	-0.18%
ℓind (m)	6.4302e-5	2.1132e-5	-67.14%
ℓexo (m)	4.7288e-5	5.7344e-6	-87.87%

ΔT = -5.320 K

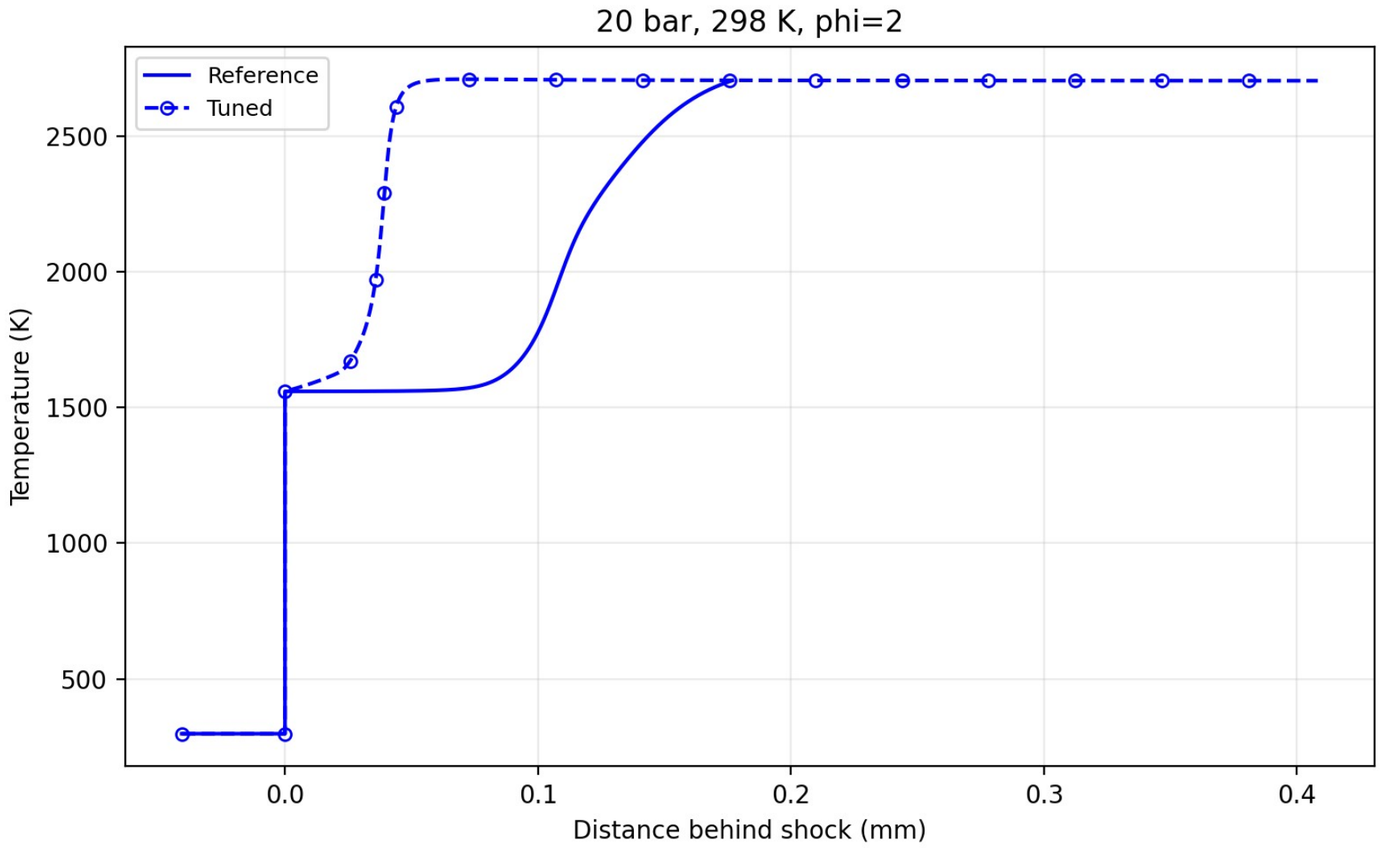
|ln(ℓind,model / ℓind,FFCM2)| = 1.11279

|ln(ℓexo,model / ℓexo,FFCM2)| = 2.10979

The tuned exothermic length is 8.2× shorter than the reference.

Case 12: 20 bar, φ = 2.0

Ethylene–air, initial temperature 298 K



Reported numerical comparison

Quantity	FFCM2	Tuned	Error
T (K)	2705.06	2704.03	-0.04%
ℓind (m)	1.1070e-4	3.9561e-5	-64.26%
ℓexo (m)	4.2647e-5	6.7873e-6	-84.09%

ΔT = -1.029 K

|ln(ℓind,model / ℓind,FFCM2)| = 1.029

|ln(ℓexo,model / ℓexo,FFCM2)| = 1.83791

The final temperature differs by only 1.029 K, while both lengths remain too short.

Remaining disagreement

The temperature profiles and length metrics identify the main fit limitations.

The tuned induction length is shorter in 10 of 12 cases.

The exothermic length is shorter in 11 of 12 cases.

At 20 bar and φ ≥ 1, exothermic lengths are 81.5–87.9% too short.

Before further tuning

Check length convergence with solver refinement and inspect thermicity profiles.
Then target induction position and reaction-zone width across all conditions.
These plots alone cannot identify which rate parameter causes the mismatch.